

CLANCY CRAWFORD

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Aerospace Engineering and Computational Mathematics student with experience in drone systems, mechanical design, numerical modeling, and structural and thermal analysis.

EDUCATION

Embry-Riddle Aeronautical University - Florida

- Bachelor of Science in Aerospace Engineering (May 2027)
Rocket Propulsion Track
- Bachelor of Science in Computational Mathematics (May 2027)
Engineering Application Track

John Wood Community College - Illinois

- Associate of Science - Engineering Science (May 2024)
Collegiate Volleyball and High Honors (3.75 GPA)
- Associate of Science: *Minor in Physics* (Dec 2023)
Collegiate Volleyball and Highest Honors (4.0 GPA)

EXPERIENCE

Mayott Aerospace —Engineer

Daytona Beach, FL | March 2026 - Present

- Evaluated materials, locking mechanisms, load distribution, and contact stresses to support heavy-lift drone payload connector.
- Designed a high-power battery cell pack and contributed to high-voltage powertrain research and development.
- Developed material comparison tools and performed preliminary sizing calculations to evaluate connector design options.
- Created presentation materials supporting Mayott Aerospace's 2nd place finish and \$10,000 award at Launch Your Venture.

PROJECTS

Cislunar Transportation Architecture – Preliminary Design

Fall 2026 - Present

- Contribute to a team developing transportation architecture for human and cargo operations throughout cislunar space.
- Current teamwork includes transportation trade studies, vehicle work breakdown, and a system requirements outline in preparation for the Systems Requirements Review.

Magnetic Alignment Dynamics for Drone Payload Docking

May 2026

- Modeled a two-axis magnetic docking connector for a heavy-lift drone using nonlinear rotational differential equation dynamics.
- Delivered a technical report with derivations and MATLAB/ODE-based modeling to support connector design development.

Avionics Box Structural & Thermal Analysis - SOLIDWORKS

Apr 2025

- Modeled support and pipe structures; delivered technical reports and recommendations on material selection and design.
- Calculated limit and ultimate loads and evaluated normal, shear, and principal stresses across multiple flight conditions.
- Conducted thermal & airflow analysis, developed performance criteria for temperature, and contributed to planning integration.

Experimental Aerodynamics & Wind Tunnel Analysis – MATLAB

Spring 2026

- Conducted wind tunnel experiments on airfoil, canard-delta wing, and supersonic nozzle configurations.
- Reduced data to calculate aerodynamic coefficients, pressure distributions, and uncertainty across tested configurations.

Causal Analysis of Aviation Accident Lethality

May 2026

- Analyzed 5,268 fatal aviation accidents using Python, converted narrative summaries to investigate high-lethality outcomes.
- Applied FCI causal discovery, conditional independence, and regression causal effect estimation to evaluate relationships.

SKILLS

Engineering Software: SOLIDWORKS, CATIA, Fusion 360, Femap, Simulink

Programming & Computing: MATLAB, Python, Java, Linux/Unix, GitHub, HTML, CSS, MobaXTerm, LaTeX

Technical: CAD, GD&T, structural and thermal analysis, wind tunnel data reduction, regression modeling

Certifications: Microsoft Office Specialist - Word, Excel, PowerPoint

RESEARCH

Orbital Debris and Collision Avoidance in Low Earth Orbit

Investigated orbital debris growth, tracking limitations, collision-risk assessment, and mitigation strategies for spacecraft in low Earth orbit; presented findings in a technical report and presentation.

LEADERSHIP & INVOLVEMENT

ERAU: Club Volleyball Treasurer, Society of Women in Engineers (SWE), Society of Women in Space Exploration (SWSE), Embry Riddle Future Space Explorers and Developers Society (ERFSEDS)

JWCC: Vice President, Phi Theta Kappa (PTK); Vice President, S.T.E.M. Club; Collegiate Volleyball Athlete, SGA

AWARDS

- Coca-Cola's Academic Team (National Level) - Gold Scholar May 2024
- All-USA Academic Team by Phi Theta Kappa Honor Society (National Level) May 2024
- Microsoft Office Specialist State Champion — Word & Excel (Office 2016) May 2022
- Renewable Women of Excellence Scholarship (\$5,000) Aug 2022